

Fig. 3: Flow-chart for edge detection

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

custom GUIs for others to use—either by running them in MATLAB or as standalone applications.

The program has been divided into three different functions for Gaussian analysis and a main function called `scanner.m`. Do not edit the functions for Gaussian analysis as those are the linkers and non-executable codes. The only executable code is `scanner.m`, which uses those functions. The flow-chart of the program is shown in Fig. 3.

Some important functions used in the program are:

Colour-map. Select a built-in colour-map with 'Property Editor.' To modify the current colour-map, use

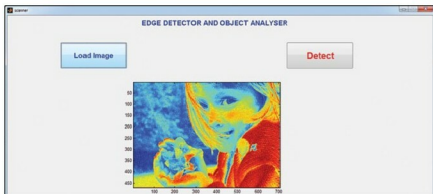


Fig. 4: Image loaded in the system

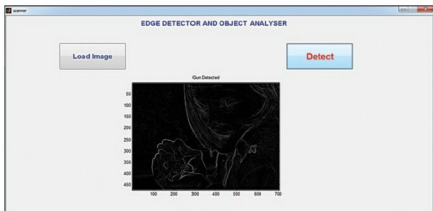


Fig. 5: Edge-detected image

'Colormap Editor,' accessible from Edit→Colormap on the figure menu.

Syntax for this function is:

```
colormap(map)
colormap('default')
cmap = colormap
colormap(ax,...)
```

conv2. This function computes the two-dimensional convolution of matrices A and B. If one of these matrices describes a two-dimensional finite impulse response (FIR) filter, the other matrix is filtered in two dimensions.

Syntax for this function is:

```
C = conv2(A,B)
C = conv2(hcol,hrow,A)
C = conv2(...,'shape')
```

Testing

To test this program, follow the below-mentioned steps:

1. Install MATLAB 7.10 or higher version in your system. Launch it from the desktop shortcut. You will see a blank command window.

2. Download the source code

folder.

3. Select any coloured picture and convert it to gif format. Save this picture as 'test' in the folder where MATLAB program is present. The same name is used in the source code. If you want to change the name of the image, you have to change it in the source code too.

4. Run the program 'scanner.m.' A graphic user interface will appear as shown in Fig. 2.

5. Click 'Load Image' button, which will load the image into the system and calculate the parameters and convert the image into jet colour format as shown in Fig. 4.

6. Now click 'Detect' button in order to detect the edges of the image you have chosen (see Fig. 5).

Thus the edges are detected successfully and you can easily characterise the various objects inside. ●

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